

MODEL QUESTION PAPER
DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
CAD/CAM

[Time : 3 hours]

(Maximum Marks: 100)

PART - A

(Maximum marks: 10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks

1. Define CAD
2. Describe rapid prototyping
3. Describe MCU in NC machine
4. Describe ATC
5. Name different types of display of CAD system

(5 x 2 = 10)

PART –B

(Maximum marks: 30)

II. Answer *any five* of the following questions. Each question carries 6 marks.

1. Explain different types of printers
2. Explain features of 3D printing
3. Explain the features of DNC
4. Explain recirculating ball and screw
5. Explain features of spindle drive
6. Write a basic configuration of workstation
7. How a machine centre is different from a machine

(5 x 6 = 30)

PART – C

(Maximum marks: 60)

(Answer *one* full question from each module. Each question carries 15 marks)

Unit – I

- | | | |
|------|---|---|
| III. | (a) With a sketch explain the working of plasma panel | 8 |
| | (b) Explain in detail two basic categories of computer memory | 7 |

OR

- IV. (a) List and give short notes on CAD software packages 7
 (b) Explain computer networking 8

Unit – II

- V. (a) List the applications of CAM 8
 (b) Identify the areas where NC technology can be applied 7

OR

- VI. (a) Explain the advantages of CAPP 7
 (b) With help of block diagram explain sequential engineering 8

Unit – III

- VII. (a) Explain contour NC machine and its features 7
 (b) Describe features of DNC 8

OR

- VIII. (a) List the advantages of higher axes machines 7
 (b) Explain the axes of Lathe and vertical milling machine 8

Unit – IV

- IX. (a) Explain a block of NC program 8
 (b) Explain different tool offset 7

OR

- X. Write a CNC part program for milling a rectangular block.

Assumptions: (i) material removed in one cycle of operation (ii) top surface of the workpiece is $z = 0$ (iii) thickness of material is 2 mm (iv) end mill diameter 5mm (v) other data required may be suitably assumed

(Explanation is to be provided on right side of each block of program) 15

All dimensions are in mm

Portion where material to be removed entirely

